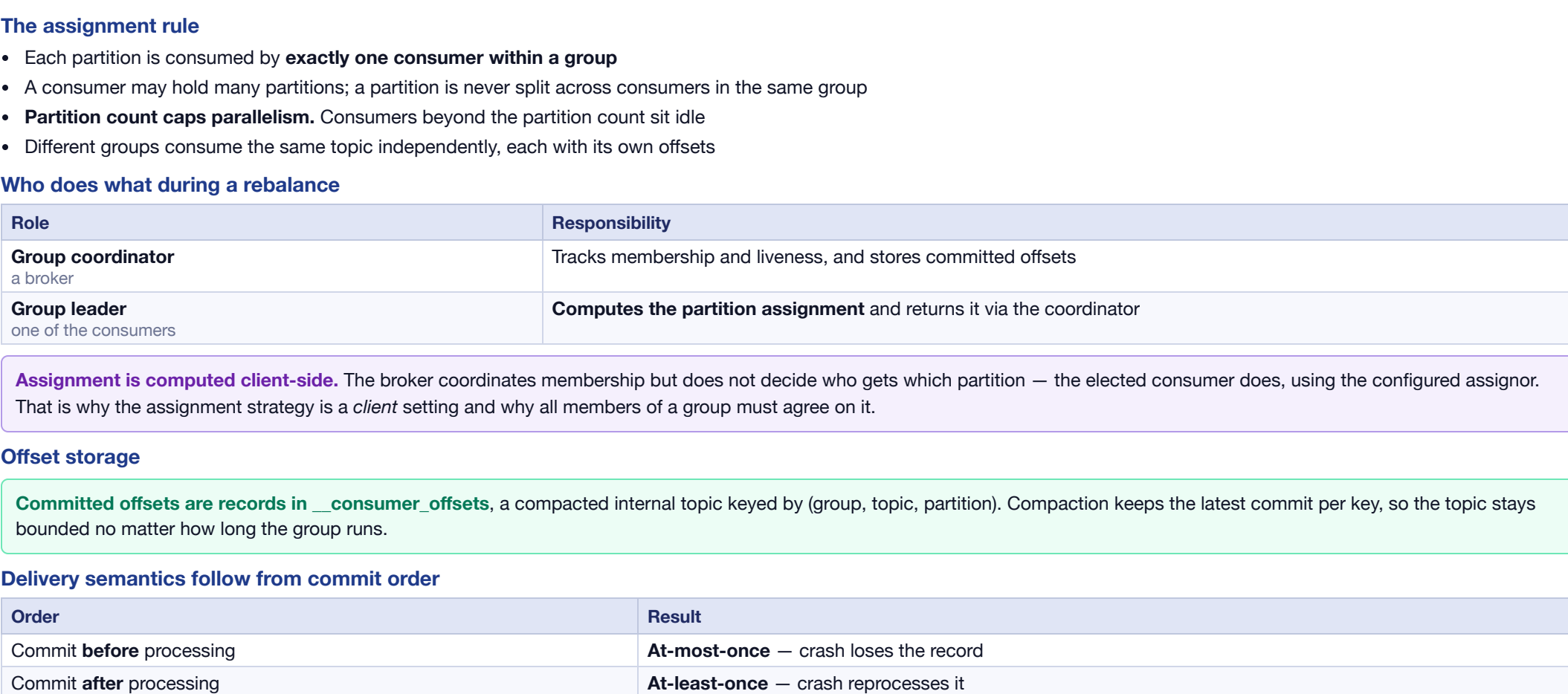
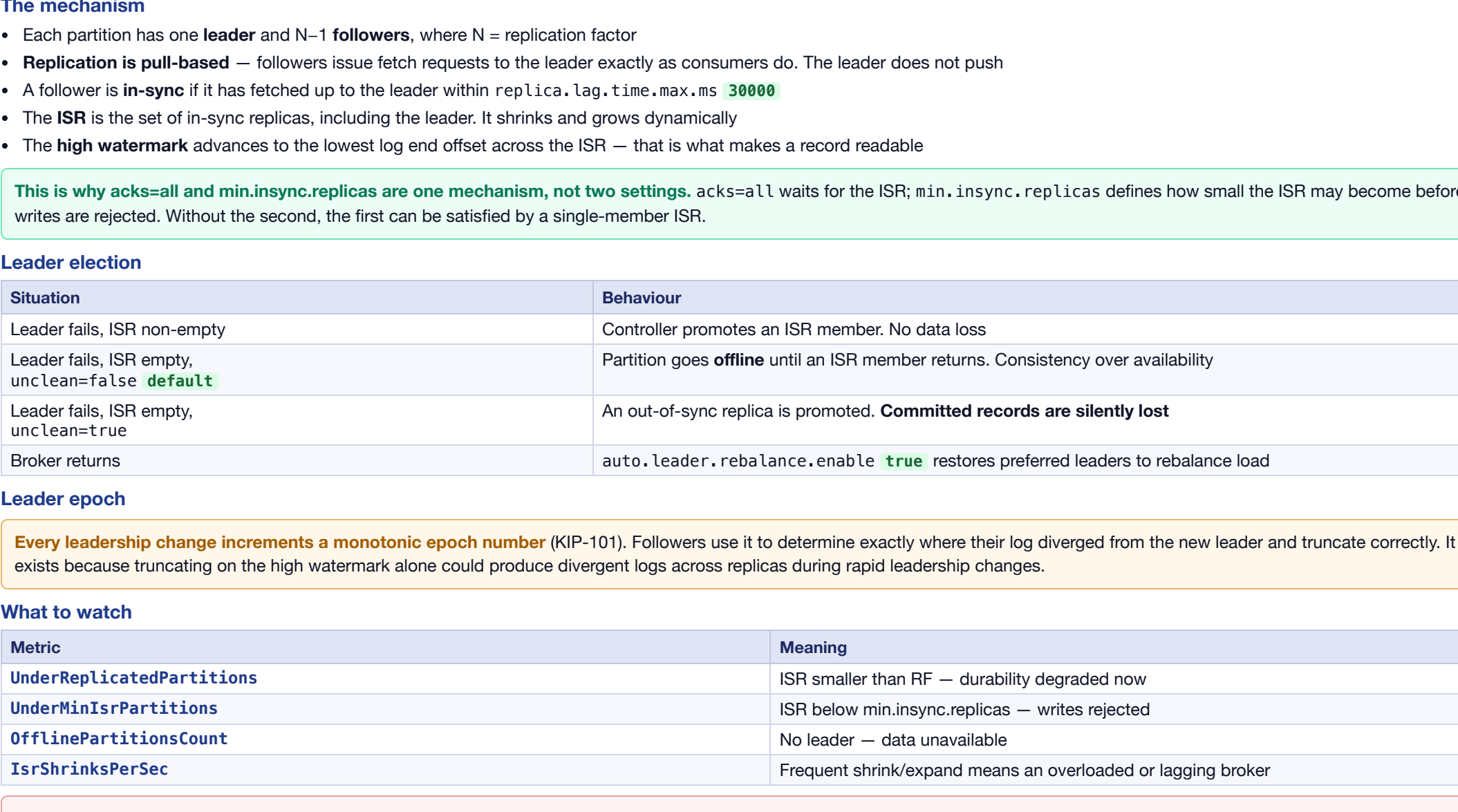
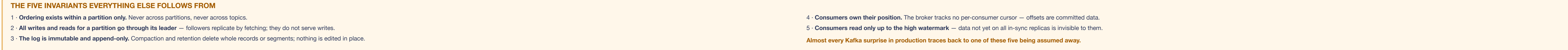
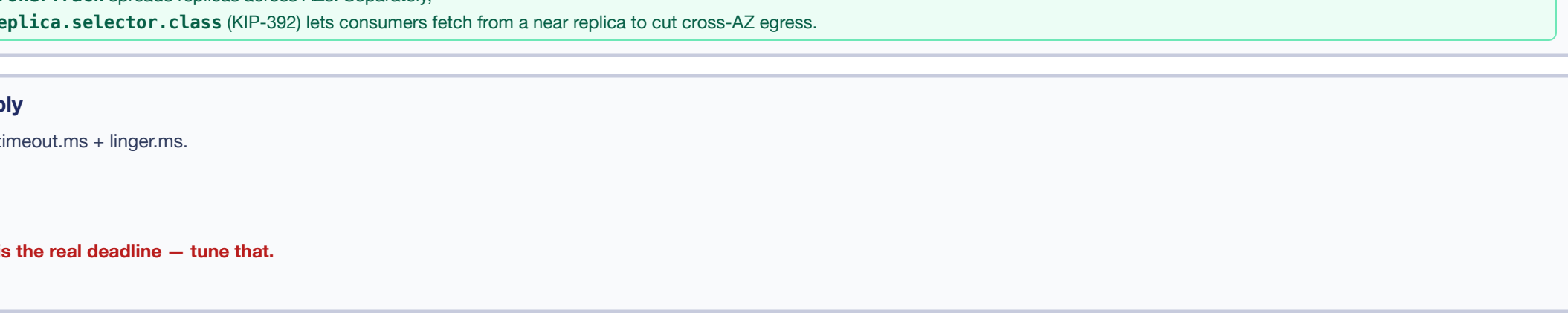


brokers, the controller quorum, partitions and the internal topics that make it work.



three settings decide whether you can lose data — and they only work together

at each combination gives you



9 · TOPIC & BROKER CONFIG

Internal topics need production settings too. `offsets.topic.replication.factor` and `transaction.state.log.replication.factor` default to 3, but single-broker dev setups often lower them — and that value persists if the topic was created before the cluster grew.

<pre> true none / 1s4 10384 0-2 - 3 2 false delete hours=1 d small true read_uncommitted latest 1-50 at_least_once entity_id </pre>	<pre> true (simplified) 1s4 23768 5-10 required 3 2 false delete 7 d default false (100) read_committed earliest 100-500 exactly_once_v2 join key </pre>
---	--

Starting points, not answers. Batch, linger and poll values depend on message size, throughput and hardware. Take the nearest column, load-test with your payloads, then change one variable at a time. **Total**

14 · SECURITY, QUOTAS & SIZING

parameter	Values
security.protocol	PLAINTEXT, SSL, SASL_PLAINTEXT, SASL_SSL
sasl.mechanism	PLAIN, SCRAM-SHA-256, SCRAM-SHA-512, GSSAPI, OAUTHBEARER
sasl.enabled.authentication	https -> empty disables hostname verification. Do not
sasl.algorithm	
sasl.client.auth	none - required (NTL5)

PLAINTEXT is the default: no encryption, no authentication. Any client that can reach the port can read and write every topic. Note also that **TLS disables the zero-copy path**, so budget CPU time - but do enable it.

ACLs and quotas

- Producers need **Write** on the topic; transactional producers also need **Write on the Transactional**
- Consumers need **Read** on the topic and on the consumer group
- producer_byte_rate / consumer_byte_rate / request_percentage **throttle rather than reject** - a misbehaving client slows down instead of taking the cluster with it

Sizing

```

partitions = (daily_data_volume
/ partitions_a_throughput / per_partition_rate
/ producers_per_peak_consumer_instances)

bytes = rate * avg_size * retention_s * RF
= compression_ratio * size * retention_s * RF

```

More partitions is expensive. Every costs file handles, memory on brokers and clients, and controller work. Over-partitioning raises latency and lengthens recovery after a broker failure.

Alert on lag trend, not a fixed threshold. Steady lag under load is normal; monotonically growing lag means consumption is losing to production and will not recover on its own.

- The test that validates the whole sheet.** Take one broker out under production-like load. Writes should continue, no data should be lost, consumer lag should recover, and no alert should be a surprise. **Configuration never tested against a real failure is a hypothesis** — and the failure is the wrong moment to find out which setting you got wrong.